Reinforcement Learning for Logic Synthesis
Library Tuning for Technology Mapping

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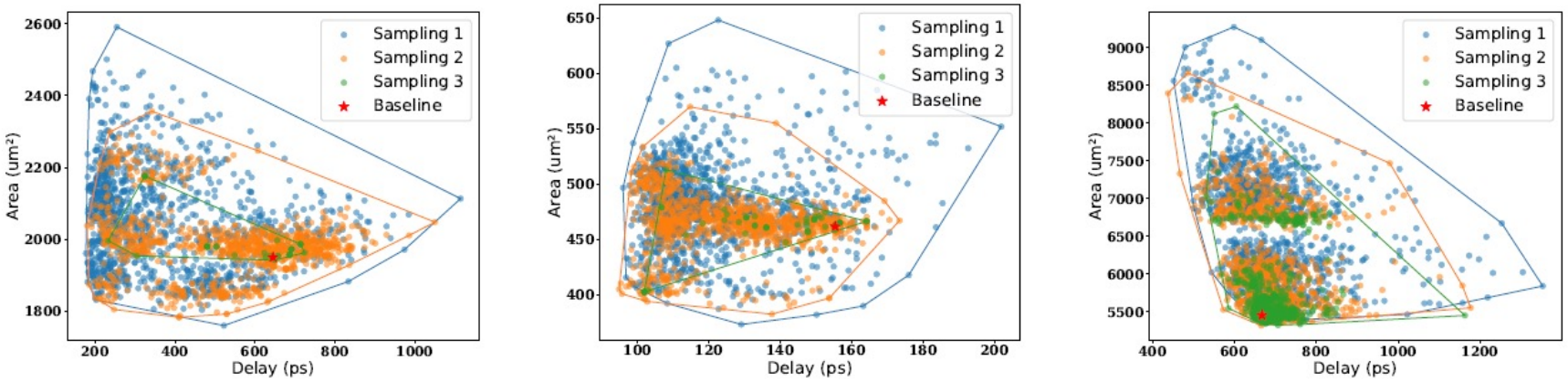
M. Liu, D. Robinson, Y. Li and C. Yu, "MapTune: Advancing ASIC Technology Mapping via Reinforcement Learning Guided Library Tuning," *2024 ACM/IEEE International Conference On Computer Aided Design (ICCAD)*, Newark, NJ, USA, 2024, pp. 1-10.

Introduction

- Technology mapping is a critical phase in the logic synthesis process, converting high-level circuit descriptions, such as those at the Register Transfer Level (RTL), into technology-specific gate-level netlists
- This step not only follows logic optimization but is also essential for optimizing Power, Performance, and Area (PPA)
- Technology libraries are critical for technology mapping, as they provide pre-defined gates and components optimized for specific fabrication processes

Introduction

- **MapTune observes and demonstrates a counter-intuitive finding that partially selected set of cells from the full technology library might significantly improve the technology mapping PPAs.**



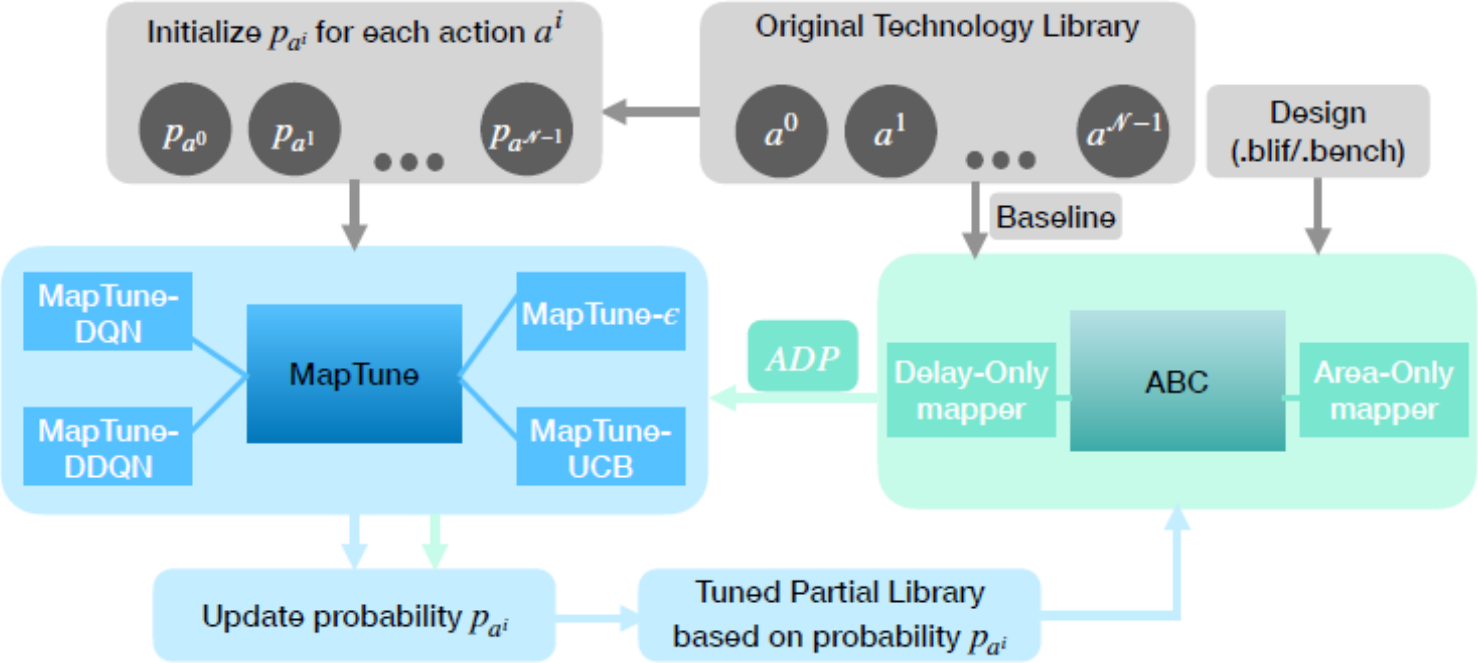
(a) s13207 (b) c2670 (c) b20

Figure 1: Technology mapping results of selected designs: Baseline: All 161 cells of ASAP7 library; Sampling 1: Randomly sampling 75 - 100 cells; Sampling 2: Randomly sampling 100 - 125 cells; Sampling 3: Randomly sampling 125 - 150 cells.

https://dl.acm.org/doi/pdf/10.1145/3676536.3676762

MapTune

- **Reinforcement learning for library tuning (cell selection)**
  - **In each iteration, a subset of *n* candidates is selected and then the reward (ADP) is computed**



RL Formulation

- **Action space**
  - Define each action as a^i, taking action a^i means selecting cell i
  - Cardinality = N
- **State**
  - State $S = [S_0, S_1, \ldots, S_{N-1}]$, where $S_i \in [0, 1]$ means cell i is selected or not
  - When $\sum^{(N-1)}_{(i=0)} S_i = n$, it refers to a complete state

RL Formulation

- **Reward**

$$\mathcal{R}_S = -ADP_S = -\left(\frac{D_S}{D_{Base}} \cdot \frac{A_S}{A_{Base}}\right)$$

  - D_S and A_S are the metrics derived from the technology mapping of the design using the selected subset of cells indicated at state S
  - D_Base and A_Base represent the baseline metrics, established using all the cells

- **Probability vector** ***P*= [*P*_a^0, *P*_a^1, ..., *P*_a^(N−1)]**
  - ***P*_a^i is defined as the likelihood of selecting cell i for the sampled library, which can maximize the reward**

Install MapTune

- git clone https://github.com/Yu-Maryland/MapTune.git
- cd /Maptune
- pip3 install torch torchvision --index-url https://download.pytorch.org/whl/cpu
- pip install gymnasium
- export PATH=/your/path/to/ABC:${PATH}

https://github.com/Yu-Maryland/MapTune

Libraries

- **5 .genlib**

.genlib	# of gates
7nm.genlib	161
gf180mcu_ff_125C.genlib	151
gf180mcu_tt_025C.genlib	151
nan45.genlib	94
sky130.genlib	343

Assignments

- **Assignment 1 (30%): Environment setup and baseline replication**
  - Install the MapTune framework and establish baseline metrics
  - For every library, conduct a baseline technology mapping using the full cell set for every design
  - Record the baseline Area and Delay
  - ABC Tip: read library.lib; map; topo; upsize; dnsize; stime
- **Assignment 2 (30%): Randomly sampling replication**
  - Validate the finding that a partial library can outperform a full library
  - Perform random sampling of ASAP7.genlib at three levels: 75-100 cells, 100-125 cells, and 125-150 cells
  - Plot the results on an Area-Delay scatter plot, similar to Figure 1 in the slides for the three benchmarks
    - s13207, c2670, and b20

Assignments

- **Assignment 3 (40%): Standing on the shoulders of MapTune**
  - **Develop an automated optimization method to identify a cell subset (smaller than or equal to the MapTune baseline) that achieves a superior Area-Delay Product (ADP) across various design constraints**
  - **Baseline of MapTune**
    - **Benchmarks: s13207, c2670, and b20**
    - **Libraries: nan45.genlib and sky130.genlib**
    - **Enforce fixed cell selection budgets at 30%, 50%, and 70% of the total library cardinality**
    - **Find the optimal subset within these fixed budgets as the baseline**
      - The subset with the minimum ADP
  - **Develop and implement an automated method to outperform the MapTune results**
    - **Your method must find a subset where $|S_{new}| \leq |S_{MapTune}|$ and $ADP_{new} < ADP_{MapTune}$**
    - **$|S_{MapTune}|$ and $ADP_{MapTune}$ can be the inputs to your method**

Delivery & Due Date

- Delivery
  - Report summarizing your experimental results (table and analysis)
  - Others: to be announced by TA

- Due on 2026/6/1 before class starts